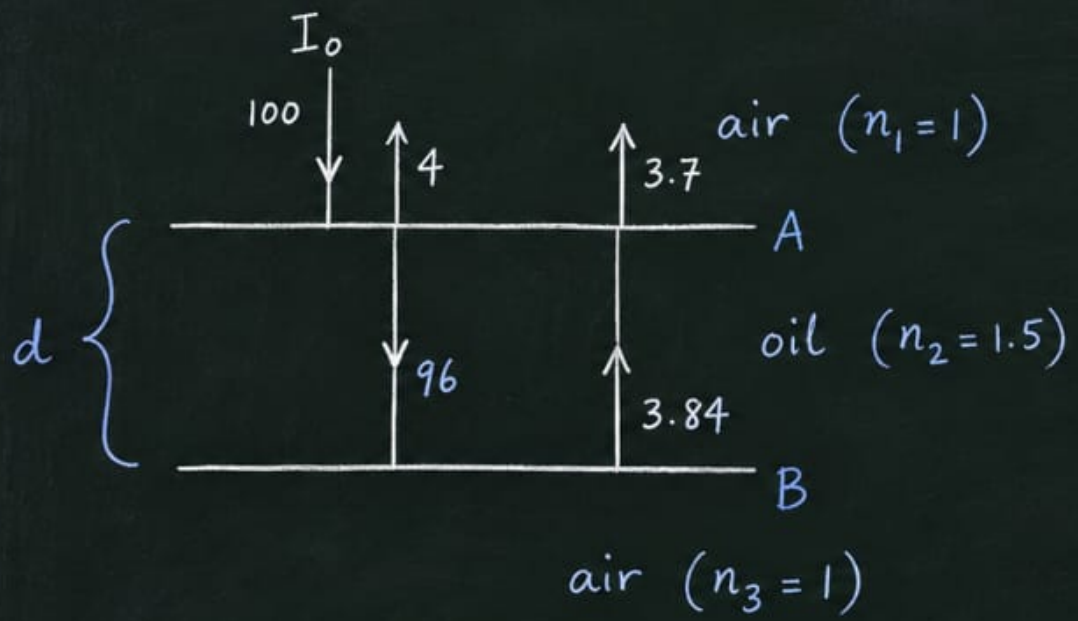
Thin film Interference



Reflectivity

$$r_A = \frac{n_1 - n_2}{n_1 + n_2} = -0.2$$

This is 4% of I_0.
(Intensity is square of amplitude)

$$r_B = \frac{n_2 - n_3}{n_2 + n_3} = 0.2$$

which is 4% again.

4% of 96% is 3.84

Therefore Intensity reflected up from B is 3.84% of I_0

Question !: If you look from above, what can the 2 waves constructively or destructively interfere?

Difference in path $= 2d$

Difference in phase $= \dfrac{2\pi}{\lambda_{oil}} 2d = \dfrac{4\pi d}{\lambda_{oil}}$

There is an extra phase of π coming from the (-) sign of the wave reflected from the top surface. Thus,

$$\text{total phase difference} = \frac{4\pi d\, n_2}{\lambda_{air}} + \pi$$

Remember, if n_3 is made larger than n_2, the "π" will disappear.

Destructive interference :

$$\frac{4\pi d\, n_2}{\lambda_{air}} + \pi = (2m-1)\pi \qquad m = 1, 2, 3, \ldots$$

Constructive interference :

$$\frac{4\pi d\, n_2}{\lambda_{air}} + \pi = 2m\pi \qquad m = 1, 2, 3, \ldots$$

If you make $d \to 0$ what happens ?

for example $\frac{d}{\lambda_{oil}} << 1$ then destructive !

Example (Let us look at destructive interference: what d gives destructive ?)

$\lambda_{air} = 400$ nm (Blue) $\qquad \lambda_{oil} = \frac{400}{1.5} = 267$ nm

$m = 1 \qquad d = 0$

$m = 2 \qquad d = 133$ nm

$m = 3 \qquad d = 267$ nm

If you send white light into a film of thickness $d = 267$ nm. (We know this gives destructive interference for blue light from last slide)

	λ_{air} (nm)	$k\delta$	$\cos^2 \frac{k\delta}{2}$
Blue	400	3π	0
Yellow	500	2.6π	0.35
Red	650	2.2π	0.90

(This is done by comparing with what we know from interference lecture. Total phase diff. between 2 waves is what we called '$k\delta$' there.

Two waves that are off by a phase diff of $k\delta$ will have intensity at observation point proportional to $\cos^2 \frac{k\delta}{2}$)

For this thickness, blue color will be missing from the colors you see. If your film has some other thickness, some other color might be missing.

Newton's Rings

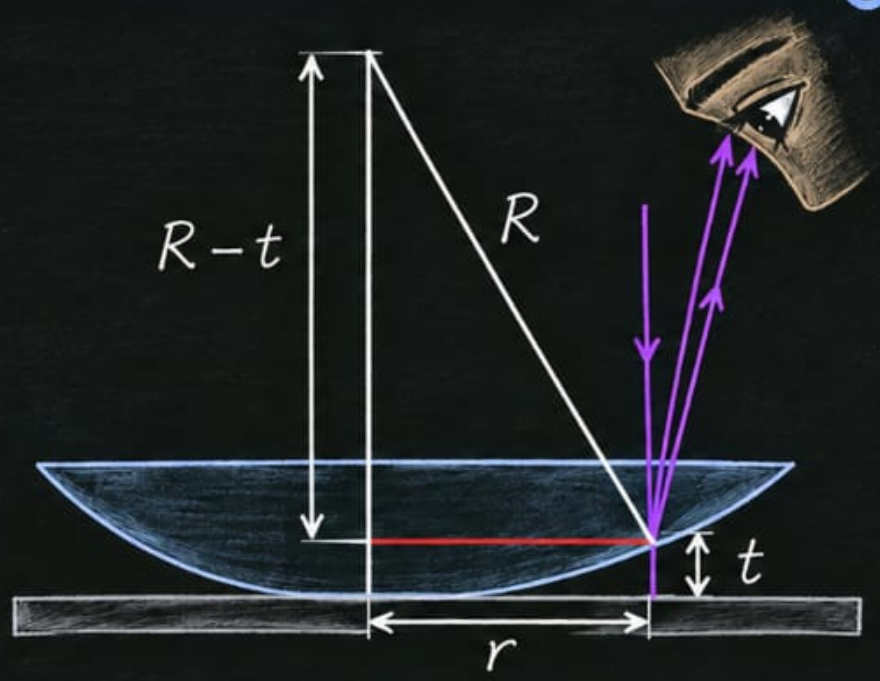


Figure 2.

$$(R-t)^2 + r^2 = R^2$$

$$r^2 = 2tR$$

t'' = varying thickness

path diff = $2t$

phase diff = $2t\left(\frac{2\pi}{\lambda}\right) + \pi$

Const. $2t\left(\frac{2\pi}{\lambda}\right) + \pi = m\,2\pi$ $\quad m = 1, 2, \ldots$

Dest. $2t\left(\frac{2\pi}{\lambda}\right) + \pi = (2m-1)\,\pi$ $\quad m = 1, 2, 3, \ldots$

Const.

$$\frac{r^2}{R}\frac{2\pi}{\lambda} + \frac{2\pi}{2} = m\,2\pi$$

$m = 1, 2, 3, \cdots$

$$\frac{r^2}{R} = \left(m - \frac{1}{2}\right)\lambda$$

$m = 1, 2, 3, \cdots$

or

$$\boxed{\frac{r^2}{R} = \left(m + \frac{1}{2}\right)\lambda}$$

$m = 0, 1, 2, \cdots$

Dest.

$$\frac{r^2}{R}\frac{2\pi}{\lambda} + \frac{2\pi}{2} = \frac{2m-1}{2}\lambda$$

$m = 1, 2, 3, \cdots$

$$\frac{r^2}{R} = (m-1)\lambda$$

$m = 1, 2, 3, \cdots$

or

$$\boxed{\frac{r^2}{R} = m\lambda}$$

$m = 0, 1, 2, \cdots$